

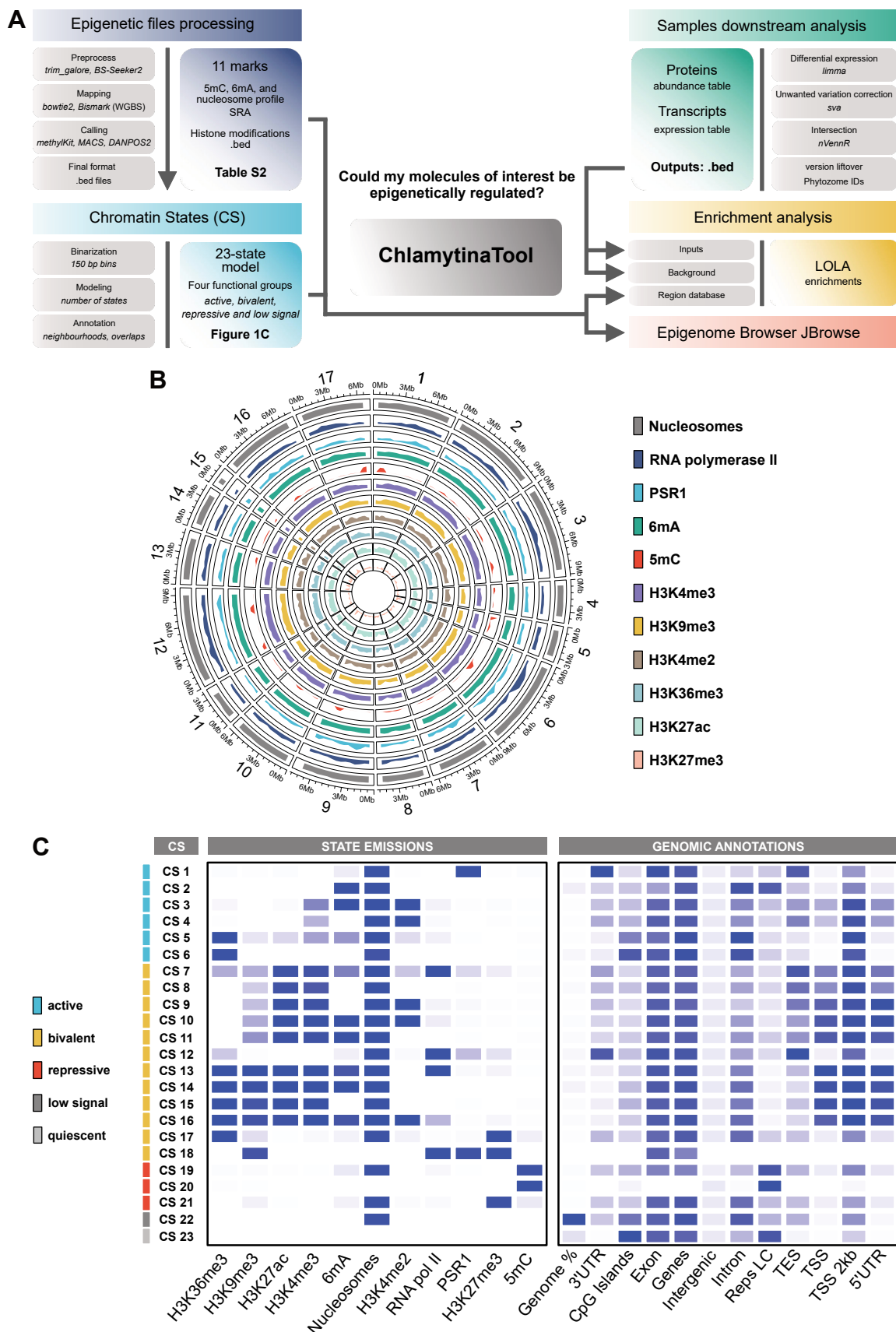


Figure 1. (A) Schematic workflow illustrating the main steps implemented in ChlamytinaTool. Input datasets corresponding to different epigenetic marks were first processed and standardized into a common .bed format. Chromatin states were then inferred through model optimization, resulting in a 23-state model that best captured the combinatorial patterns across the *Chlamydomonas* genome. The tool also integrates downstream analyses for transcriptomic and proteomic datasets, enabling multi-omic comparisons and enrichment analyses to identify potential epigenetic regulatory events. Finally, an embedded epigenome browser allows interactive visualization of chromatin states and individual epigenetic marks. **(B)** Circos plot representing the genome-wide distribution of the epigenetic marks included in ChlamytinaTool, mapped across the chromosomes of *Chlamydomonas reinhardtii*. **(C)** Summary of the 23 inferred chromatin states (CS). The heatmap shows the emission probabilities of each epigenetic mark per state, together with the relative enrichment of each state in different genomic features. Based on the combinatorial patterns of marks, CS were functionally classified as active, bivalent, repressive, low signal, or quiescent.

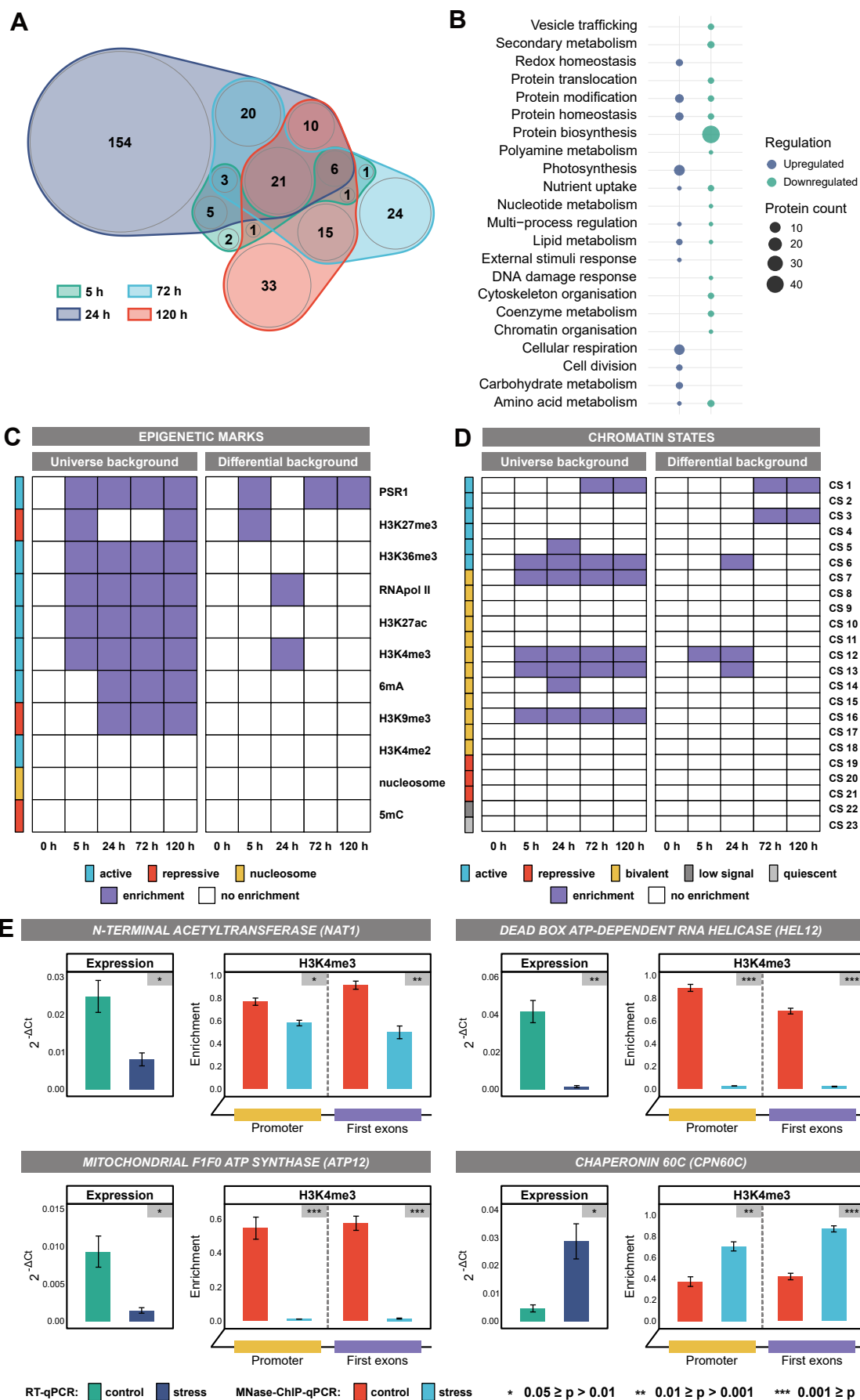


Figure 2. (A) Quasi-proportional Venn diagram showing the number differentially expressed proteins between control and combined heat and osmotic stress across the different experimental time points. (B) Functional enrichment analysis of exclusive proteins regulated at 24 h, categorized according to MapMan functional bins. This representation highlights the main biological processes and metabolic pathways affected during the stress response. (C) Enrichment of *Chlamydomonas*' proteome under combined heat and osmotic stress in different epigenetic marks and (D) in the 23 chromatin states generated. Universe refers to using the whole proteome as background, while differential refers to using only the differentially expressed proteins as background. (E) Validation of candidate genes identified through integrative analysis. Expression levels of *NAT1*, *HEL12*, *ATP12*, and *CPN60C* were quantified by RT-qPCR under control and stress conditions. Enrichment of the activating histone mark H3K4me3 in promoter and first exon regions of these genes was assessed by MNase-ChIP-qPCR, confirming their epigenetic regulation under combined stress.